Can Power Draw Constrain Covert Compute?

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TL;DR

How much compute can an operator hide from a power meter? We derive the answer in closed form and measure every input on real GPUs: up to 1.16 machines' worth, falling to 0.07 under reasonable assumptions.

CONTRIBUTIONS

1. A closed-form floor β: the largest covert computation a power trace cannot rule out, in machines of declared capacity.
2. Every input of the closed form measured on NVIDIA A100s — band edges, overhead, and the marginal cost of covert work.
3. The floor evaluated end-to-end: the trace concedes $\beta^* = 1.16$ machines — more than one whole machine of ambiguity; cheating runs we execute attain 0.41 of it.
4. A ladder of verifier capabilities and assumptions that shrink the floor — requiring covert work to be *useful* pushes β down to 0.07.

READ MORE

Paper & code: github.com/tomkimpson/power-to-flops
Companion: github.com/tomkimpson/detecting-training-from-power-traces
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Read more

Motivation

- International agreements on frontier AI require *verification*: an outside party must be able to check what an operator actually ran, even when that operator may cheat and try to hide their computation.
- Hardware mechanisms for verification typically rely on reading silicon the operator controls.
- Off-chip analogue sensors (e.g. a power meter) have been identified as a potential verification channel that is beyond the operator's reach.
- We ask the question: given only the power trace, **how much computation can the operator hide?**

From power to FLOPs?

- A power meter sees only $P(t) = r(t)\,F(t) + P_0(t)$: an operation rate F priced at an exchange rate r [J/op], on top of an overhead draw P_0.
- The rate r is not a device constant — it varies with how the machine is run.
- Given $P(t)$ (or its integral $E = \int P(t)dt$) we want to know $F(t)$ (or its integral $C = \int F dt$); an inverse problem.
- **BUT** the equation is non-injective and so cannot be inverted.
- We instead bound the width of the covert compute interval, $\beta \propto C_{\text{hi}} - C_{\text{lo}}$.

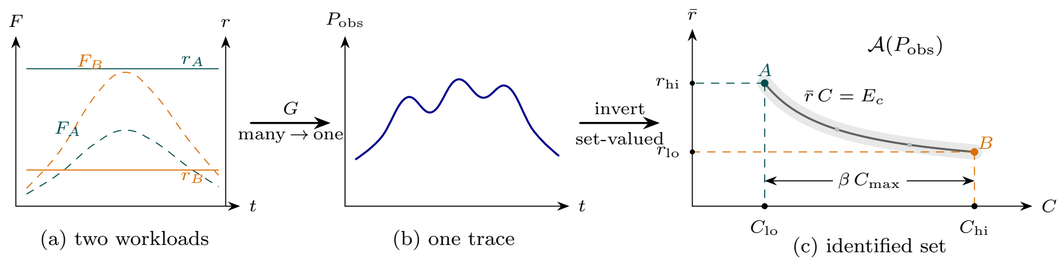


Figure 1. Two workloads whose products rF coincide drive one trace; inverting recovers only the interval $[C_{\text{lo}}, C_{\text{hi}}]$, whose width relative to capacity is the floor β.

- The dynamic power law factorises $r = \alpha\,\kappa\,C_{\text{eff}}\,V^2$ — operands, format, locality, voltage: all chosen by the prover.
- The overhead P_0 is prover-chosen too: idled services free joules to cover covert work.
- Together these are the prover's degrees of freedom: declare work at expensive settings, run it cheap, and spend the saved joules on covert compute.

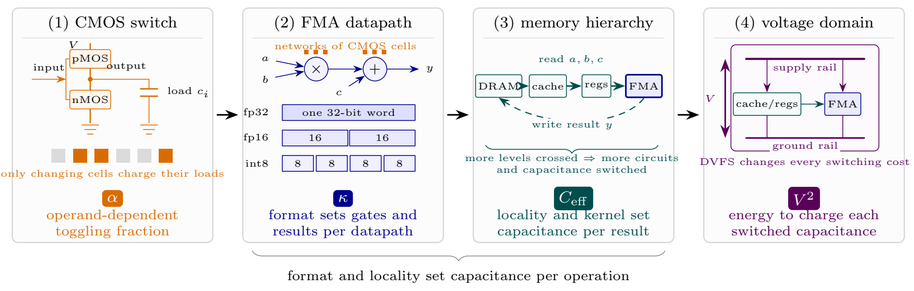


Figure 2. The exchange rate assembled across physical scales, from a single CMOS switch to the powered die.

Experiments: measuring the hiding room

- Matmul sweep (format × operands × locality): declared work prices anywhere from 19.5 down to 0.786 pJ/op — a 24.8× span, reproducing across four A100s to 2–3%.
- Idle-state sweep: the overhead P_0 ranges over 63–96 W — joules a prover frees by idling services.
- Covert dosing: covert work added to a declared burst costs 0.51 pJ/op at the margin for synthetic matmuls, rising to 2.5–3.2 pJ/op for a real transformer block (Pythia-6.9B).

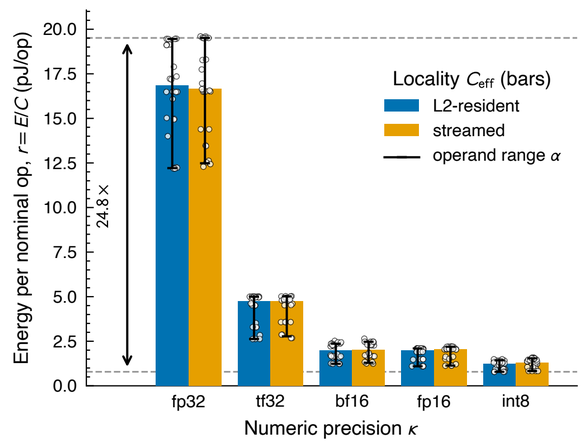


Figure 3. The measured energy quotient E/C, swept factorially over precision (bar groups), locality (paired bars) and operand distribution (whiskers); dashed lines mark the band edges.

Pushing down β

- Even with the declared precision checked, the trace concedes $\beta^* =$**1.16 machines** — more than one whole machine of ambiguity. Matched-energy cheating runs attain 0.41 of it, with every run accepted by the meter.
- Additional verifier capabilities push it down, e.g. re-executing the declared work at an observed operating point: $\beta \to$ **0.34**.
- Reasonable assumptions push it down further, e.g. assuming the adversary needs *useful* covert work (multiplying zero matrices is cheap and easy to hide, but useless): $\beta \to$ **0.07**.

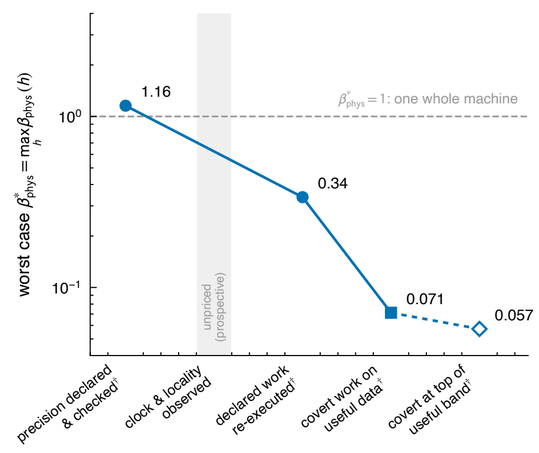


Figure 4. The capability ladder: worst-case covert compute β^* (machines of declared capacity) as the verifier gains capabilities — model upper bounds, whereas the 0.41 in the text is what our cheating runs actually attained. The dashed line marks one whole machine of conceded ambiguity.

Limitations & implications

Limitations. One GPU model measured • voltage axis unswept • power read from on-die NVML telemetry, not an off-chip meter (±5% adopted) • the floor prices a cheating corner, so it is an upper bound.

Implications. Power traces are a viable but limited channel for compute verification • governance protocols should treat power as one layer in a suite of verification options.

Companion work. Reading time-resolved power rather than its integral, the same meter *can* detect the cadence of synchronous training (WIP, repository in sidebar).